

BOSONIZATION OF THE KONDO PROBLEM

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We present a bosonized action for the $SU(N)$ Kondo problem within the framework of path-integral non-abelian bosonization. We show that the model is equivalent to a level $k = 2N + 1$ Wess–Zumino–Witten theory coupled to impurity fields. We show that the system is scale invariant in the strong-coupling limit by explicit computation of the β -function. We calculate the magnetic susceptibility and the low-temperature specific heat in the same limit.

1. Introduction

In this paper we apply the method of non-abelian bosonization, in its path-integral form, to the Kondo problem. This model has been extensively investigated through a variety of methods. Anderson et al. [1] produced the first path-integral study of this problem and mapped it into a one-dimensional Ising model with $1/r^2$ interactions. Anderson [2] studied this problem in a pioneering paper using what now we know as renormalization-group methods. Wilson [3] presented a numerical solution of this problem, exhibiting for the first time the famous Kondo crossover. Later, an exact solution was found using the Bethe-ansatz technology [4, 5]. The Bethe ansatz yielded an exact solution for both the energy spectrum and the thermodynamics. Finally, Read and Newns [6] developed a large- N approach within a path-integral framework for the Kondo model, or rather the Coqblin–Schrieffer model [7].

It is rather remarkable, however, that the methods of bosonization have not been extensively applied to this problem. The bosonization technique has been highly

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useful in the framework of one-dimensional quantum Fermi systems and field theories [8–12]. It has been realized that the one-impurity Kondo problem is a one-dimensional quantum problem because only the s-wave contributes to the scattering amplitude. It is this one-dimensional character what makes the Bethe ansatz, and indeed even the Wilson's numerical approach, work. Given the non-abelian symmetry of the problem ($SU(N)$ for the Coqblin–Schrieffer model) it is tempting to use the method of non-abelian bosonization developed by Witten [13].

However, the Kondo problem is not precisely one-dimensional in the following sense. In Wiegmann's approach [5] the Kondo model lives on a half-line because it is a *radial* problem. Thus, the Fermi fields have both “left-” and “right-” moving components and a “hard” boundary condition at $r = 0$ where the scattering off the impurity takes place. Andrei [4], in contrast, describes the problem on a whole line with *only* right movers (or left movers) and with the impurity somewhere on the line. This approach, which corresponds to the representation that we use, has the difficulty that it is a chiral–Fermi theory. In other words, it does not have left–right symmetry. Thus, the quantization of this theory has the ambiguities inherent to all theories of chiral fermions. These issues are side stepped by the Bethe ansatz in ways which are not understood for the present time.

The path-integral approach focuses precisely on these difficulties. In fact, the key to the path-integral approach to bosonization is the careful treatment of the chiral anomaly [14,15]. Fortunately, recent advances in the treatment of chiral gauge theories [16] have yielded a way to deal with such problems.

In this paper we present a bosonized action for the Kondo model using the methods of the path-integral non-abelian bosonization. Our bosonized action describes right-moving Fermi fields (the band electrons) interacting with a localized magnetic impurity. We map this problem into a theory of a Wess–Zumino–Witten bose field coupled to another boson representing the impurity. We present three different descriptions of the impurity. Two are based on a local constrained Fermi field and in the last we use geometric quantization. In the descriptions with a constrained fermion we use (a) an analytic continuation of the Wess–Zumino–Witten theory and (b) a temporal Wilson loop to deal with the constraint. Our main results are the bosonized actions. We also discuss the strong-coupling limit of this model and show that it is a scale-invariant theory representing a completely screened impurity. We calculate the β -function at strong coupling and show that it vanishes as the antiferromagnetic exchange constant g (the coupling constant) goes to infinity. We also calculate the magnetic susceptibility and the low-temperature specific heat at $g = \infty$.

The paper is organized as follows. In sect. 2 the model is presented, non-abelian bosonization techniques are used and the bosonized actions are obtained. In sect. 3 we calculate the β -function at strong coupling. Sect. 4 is devoted to the study of magnetic susceptibility and low-temperature specific heat, both in the strong cou-

pling limit. Sect. 5 is devoted to the conclusions. Three appendices contain details of the calculations.

2. Bosonization of the Kondo model

We shall consider the following two-dimensional lagrangian (in Minkowski space) for the Kondo model

$$\mathcal{L}_k = i\psi^\dagger \partial_+ \psi + id^\dagger \partial_t d - g\psi^\dagger t^a \psi d^\dagger t^a d, \quad (2.1)$$

where ψ is a left-handed fermion representing the electron

$$\psi = \begin{pmatrix} 0 \\ \psi_L \end{pmatrix}, \quad (2.2)$$

and d is a right-handed fermion representing the impurity,

$$d = \begin{pmatrix} d_R \\ 0 \end{pmatrix}. \quad (2.3)$$

In our conventions $\partial_+ = \partial_t + \partial_x$ and t^a are the $SU(N)$ generators, normalized according to

$$\begin{aligned} [t^a, t^b] &= if^{abc}t^c, \\ \text{tr } t^a t^b &= \tfrac{1}{2}\delta^{ab}, \\ f^{abc}f^{a'bc} &= \mathbb{C}(G)\delta^{aa'}, \end{aligned} \quad (2.4)$$

with $\mathbb{C}(G)$ the quadratic Casimir operator for $SU(N)$, $\mathbb{C}(G) = N$.

In the lagrangian (2.1), the (fixed) impurity interacts with the electric field through a quadratic interaction which can be eliminated using the identity

$$\begin{aligned} &\exp \left[-ig \int d^2x \psi^\dagger t^a \psi d^\dagger t^a d \right] \\ &= \int \mathcal{D}A_+^a \mathcal{D}A_-^a \exp \left[i \int d^2x (A_+^a A_-^a + \sqrt{g} A_-^a d^\dagger t^a d + \sqrt{g} A_+^a \psi^\dagger t^a \psi) \right]. \end{aligned} \quad (2.5)$$

The auxiliary fields $A_\pm$ take values in the Lie algebra of $SU(N)$ and are written as the $\pm$ components of a vector field. Actually, A_- acts as the time component of a vector since the impurity is fixed (and Lorentz invariance does not hold).

Using eq. (2.5) the generating functional reads

$$Z = \int \mathcal{D}\psi^\dagger \mathcal{D}\psi \mathcal{D}d^\dagger \mathcal{D}d \mathcal{D}A_+ \mathcal{D}A_- \mathcal{D}a e^{i \int d^2x \mathcal{L}_{\text{eff}}}, \quad (2.6)$$

with

$$\mathcal{L}_{\text{eff}} = \psi^\dagger (i\partial_+ + \sqrt{g}A_+ + F) \psi + d^\dagger (i\partial_t + \sqrt{g}A_- + H) d + a(x)(d^\dagger d - n). \quad (2.7)$$

We have introduced the Lagrange multiplier $a(x)$ in order to enforce the condition

$$d^\dagger d = n, \quad (2.8)$$

fixing the impurity number to n . F and H are external sources which can be shifted to the A_+A_- term by making the following change of variables

$$\begin{aligned} \sqrt{g}A_- + H &\rightarrow A_-, \\ \sqrt{g}A_+ + F &\rightarrow A_+, \end{aligned} \quad (2.9)$$

in eq. (2.6).

We now proceed to bosonize the Kondo lagrangian within the path-integral framework. As we stated in sect. 1, we shall follow three alternative approaches.

2.1. STANDARD PATH-INTEGRAL BOSONIZATION

In the path-integral framework, bosonization can be implemented by integrating out the fermion variables and writing the resulting determinants in terms of boson fields [13–15]. One starts from the generating functional in the form

$$\begin{aligned} Z_A &= \int \mathcal{D}A_+ \mathcal{D}A_- \mathcal{D}a \det_L(i\partial_+ + A_+) \\ &\times \det_R(i\partial_t + A_- + a) \exp \left[i \int d^2x \left(\frac{1}{g} (A_+ + F)(A_- + H) - an \right) \right]. \end{aligned} \quad (2.10)$$

With $\det_L(\det_R)$ we indicate fermion determinants for left-(right-)handed fermions. Now, as it is well known, Weyl fermion determinants are ill-defined quantities since the corresponding Dirac operator does not define an eigenvalue problem [16,17]. One usually overcomes this problem by adding free right (left)-handed free fermions. One then has

$$\det_L(i\partial_+ + A_+) \equiv \det(i\cancel{\partial} + \tfrac{1}{2}\cancel{A}(1 + \gamma_5)), \quad (2.11)$$

$$\det_R(i\partial_t + A_- + a) \equiv \det(i\cancel{\partial} + \tfrac{1}{2}(\cancel{A} + \cancel{d})(1 + \gamma_5)). \quad (2.12)$$

In eq. (2.12) analytic continuation $t \rightarrow t - x$ is assumed in the r.h.s. (see below). Now, if one uses the identity

$$\left[i\partial + \frac{1}{2}(\not{A} + \not{d})(1 + \gamma_5) \right] = e^{-(1-\gamma_5)\phi/2} \left[i\partial + \frac{1}{2}\not{A}(1 + \gamma_5) \right] e^{(1+\gamma_5)\phi/2}, \quad (2.13)$$

with

$$a = -\partial_t \phi, \quad (2.14)$$

eq. (2.12) can be written in the form

$$\det_R(i\partial_0 + A_- + a) = J(\phi) \det(i\partial + \frac{1}{2}\not{A}(1 + \gamma_5)), \quad (2.15)$$

with $J(\phi)$ a Fujikawa jacobian [18] which can be easily evaluated [16]

$$J(\phi) = \exp\left(-\frac{i}{8\pi}\right) \int d^2x \phi \partial_x \partial_t \phi. \quad (2.16)$$

Concerning the non-abelian determinants in the r.h.s. of eqs. (2.11) and (2.15), that corresponding to the electron is given by (see appendix A)

$$\det_L(i\partial_+ + A_+) = \exp(-iW(h)), \quad (2.17)$$

with

$$A_+ = -ih^{-1} \partial_+ h, \quad h \in \text{SU}(N), \quad (2.18)$$

$$W(h) = \frac{1}{8\pi} \text{tr} \int d^2x \partial_\mu h^{-1} \partial^\mu h + \Gamma_{\text{wz}}(h), \quad (2.19)$$

$$\Gamma_{\text{wz}}(h) = \frac{1}{12\pi} \text{tr} \int_B d^3y \epsilon_{ijk} h^{-1} \partial^i h h^{-1} \partial^j h h^{-1} \partial^k h. \quad (2.20)$$

The Wess-Zumino term $\Gamma_{\text{wz}}(h)$ given by eq. (2.20) is integrated over a ball B obtained by extension of two-dimensional space-time to the cap of a three-dimensional sphere [13–15].

For the impurity, the corresponding determinant is given by

$$\det_R(i\partial_t + A_-) = \exp(-i\dot{W}(l)), \quad (2.21)$$

with

$$A_- = -il^{-1} \partial_t l, \quad l \in \text{SU}(N), \quad (2.22)$$

and $\dot{W}(l)$ is related to $W(l)$, through analytic continuation, as

$$\dot{W}(l) = \frac{1}{8\pi} \text{tr} \int d^2x \partial_x l \partial_t l^{-1} + \Gamma_{\text{wz}}(l). \quad (2.23)$$

In order to rewrite Z as a path integral over h and l we use [14,19–21] (see appendix A)

$$\mathcal{D}A_+ = \exp(-2iNW(h))\mathcal{D}h, \quad (2.24a)$$

$$\mathcal{D}A_- = \exp(-2iN\dot{W}(l))\mathcal{D}l, \quad (2.24b)$$

$$\mathcal{D}a = \mathcal{D}\phi. \quad (2.24c)$$

Putting together all these results one finally has

$$\begin{aligned} Z_A = & \int \mathcal{D}\phi \mathcal{D}h \mathcal{D}l \exp[-i(1+2N)\dot{W}(l) - i(1+2N)W(h)] \\ & \times \exp\left[\frac{i}{g} \text{tr} \int d^2x (-ih^{-1} \partial_+ h + F)(-il^{-1} \partial_l l + H)\right]. \end{aligned} \quad (2.25)$$

2.2. WILSON-LOOP APPROACH

We now proceed to an alternative treatment of the impurity part of the action by direct computation of the impurity-fermion determinant. We will also take into account exactly the effects of the single-occupancy constraint.

Thus we write, for the impurity part of the partition function

$$\begin{aligned} Z_i = & \int \mathcal{D}a \exp\left[-i \int dx dt a(x, t) n(x, t)\right] \\ & \times \int \mathcal{D}d^\dagger \mathcal{D}d \exp\left[-i \int dx dt d^\dagger (i\partial_t + A_- + a) d\right], \end{aligned} \quad (2.26)$$

where

$$n(x, t) = \begin{cases} 1, & \text{where there is an impurity,} \\ 0, & \text{otherwise.} \end{cases} \quad (2.27)$$

Hence, we find

$$Z_i = \int \mathcal{D}a \exp\left[-i \int dx dt a(x, t) n(x, t)\right] \prod_x \det(i\partial_t + A_- + a). \quad (2.28)$$

This functional determinant can be evaluated quite explicitly [22,23]. The result is (see appendix B)

$$\begin{aligned} J_F = & \frac{\det(i\partial_t + A_- + a)}{\det(i\partial_t)} \\ = & \det\left[1 + \exp\left(i \int dt a(x, t)\right) \hat{T} \exp\left(i \int dt A_-\right)\right], \end{aligned} \quad (2.29)$$

where the r.h.s. is the determinant of an $N \times N$ matrix. Here A_- is in the Lie algebra of $SU(N)$ and the time-ordered exponential $U(x) = \hat{T} \exp(i \int dt A_-)$ is a group element.

Thus, we have

$$Z_i = \int \mathcal{D}a \exp\left(-i \int dx dt a(x, t) n(x, t)\right) \prod_x \det\left[1 + \exp\left(i \int dt a(x, t)\right) U(x)\right]. \quad (2.30)$$

By expanding the determinant, one readily finds that for $n = 0$, $Z_i = 1$. For $n = 1$, we get

$$Z_i(x) = \text{tr } U(x) \equiv \text{tr} \exp\left(i \int dt A_-\right). \quad (2.31)$$

We finally write

$$Z_i(x) = \prod_x \left[\text{tr} \hat{T} \exp\left(i \int dt A_-(x, t)\right) (1 - n(x)) + n(x) \right]. \quad (2.32)$$

Writing A_- in terms of l (eq. (2.22)) and solving this equation for $l(x, t)$ one obtains

$$l(x, t) = l(x, 0) \hat{T} \exp\left(-i \int_0^t dt' A_-(x, t')\right). \quad (2.33)$$

Thus

$$\hat{T} \exp\left(-i \int_0^{\bar{T}} dt A_-(x, t)\right) = l(x, \bar{T}) l^{-1}(x, 0), \quad (2.34)$$

where $\bar{T}$ is the time span of the system ($\bar{T} \rightarrow \infty$). At non-zero temperature $\bar{T}$ is the inverse temperature β .

We are still left with the change in the integration measure from $\mathcal{D}A_-$ to $\mathcal{D}l$. The result is the jacobian J_A ,

$$\mathcal{D}A_- = J_A \mathcal{D}l, \quad (2.35)$$

which coincides with the determinant evaluated above except for the fact that the corresponding operator is in the adjoint representation of $SU(N)$

$$J_A = \det(i \partial_t + i [A_-, \cdot]). \quad (2.36)$$

In appendix B we show that J_A can be related to J_F (defined above). For $SU(2)$ the result is $J_A = 2(J_F - 2)^2$. Note that for the conduction electrons we found a much simpler relation. In any event, the result of all these manipulations are factors of the

form

$$\prod_x J_A(l(x, \bar{T})l^{-1}(x, 0)) \text{tr } l^{-1}(x, \bar{T})l(x, 0). \quad (2.37)$$

Note that these factors act on the values of the matrix-valued field *on the boundaries* $t = \bar{T}$ and $t = 0$. One might think of such factors as averaging over boundary conditions. This fact will be important in the computation of the β -function below.

2.3. GEOMETRIC QUANTIZATION

The method of geometric quantization has been recently adapted to derive a path-integral for spin systems [24, 25]. The main idea here is to use coherent states of spin [26]. In this sense, this method, when applied to the Kondo problem, offers an alternative to the approach of Anderson et al. [1] who used Ising variables. The main advantage of geometric quantization is that it preserves the full rotational symmetry of the hamiltonian.

Following the treatment of Fradkin and Stone [25], we write the path-integral for fermions interacting with a single spin- s degree of freedom (for the $SU(2)$ case). Given $G = SU(2)$, $g_{\vec{n}} \in G$

$$g_{\vec{n}} = \exp \frac{1}{2} i \theta \vec{m} \cdot \boldsymbol{\sigma}, \quad (2.38)$$

with σ^a the Pauli matrices and

$$\vec{m} = \frac{\check{\mathbf{e}}_3 \wedge \check{\mathbf{n}}}{|\check{\mathbf{e}}_3 \wedge \check{\mathbf{n}}|}. \quad (2.39)$$

Now, given a hamiltonian $H(t)$ taking values in the Lie algebra of $SU(2)$, geometric quantization implies [24, 25]

$$\text{tr} \exp \left(i \int_0^T dt H^a(t) \sigma^a \right) = \int \mathcal{D}\check{\mathbf{n}} e^{iS}, \quad (2.40)$$

with

$$S = W_n - \frac{1}{2} \int_0^T dt H^a \check{n}^a, \quad (2.41)$$

$$\mathcal{D}\check{\mathbf{n}} = \mathcal{D}\mathbf{n} \delta(n^2 - 1), \quad (2.42)$$

$$W_n = \int_0^1 d\tau \int_0^T dt \check{\mathbf{n}} \cdot (\partial_t \check{\mathbf{n}} \wedge \partial_\tau \check{\mathbf{n}}). \quad (2.43)$$

For the Kondo model, consider the interaction between left-handed electrons ψ and

the impurity

$$\begin{aligned}\mathcal{H}_{\text{int}} &= g \int dx \psi^\dagger(x, t) \sigma^a \psi(x, t) \sigma^a \delta^{(1)}(x), \\ \mathcal{H}_{\text{int}} &= g H^a(t) \sigma^a, \quad H^a(t) = \int dt \delta^{(1)}(x) \psi^\dagger \sigma^a \psi.\end{aligned}\quad (2.44)$$

Use of normal path-integral for electrons and geometric quantization for the impurity leads to

$$Z = \int \mathcal{D}\psi^\dagger \mathcal{D}\psi \mathcal{D}\check{n} e^{iS_{\text{Kondo}}}, \quad (2.45)$$

with

$$S_{\text{Kondo}} = \int d^2x \psi^\dagger i \partial_+ \psi - \frac{1}{2} g \int d^2x \psi^\dagger \sigma^a \psi n^a \delta^{(1)}(x) + W_n. \quad (2.46)$$

Performing the fermionic path-integral, one finds

$$Z = \int \mathcal{D}\check{n} \det(i \partial_+ - \frac{1}{2} g \check{n} \sigma \delta(x)) \exp i W_n. \quad (2.47)$$

Instead of $g \delta(x)$ we shall consider a function $g(x)$ peaked around $x=0$ (for instance one can imagine a continuum of spin degrees of freedom which couples to the fermions only at $x=0$). We then have

$$\frac{1}{2} g \check{n} \cdot \sigma = i h^{-1} \partial_+ h. \quad (2.48)$$

Note that the l.h.s. of relation (2.48) depends both on x and t . One has then to include in eq. (2.41) an x -integration. The final form of Z is then

$$Z_{\text{without } \delta^{(1)}(x)} = \int \mathcal{D}\check{n} \exp \left(i \int dx W_n \right) \exp(iW(h)). \quad (2.49)$$

Here $W(h)$ is the standard Wess–Zumino–Witten action given by eqs. (2.19) and (2.20).

3. β -function

In this section we evaluate the Kondo model β -function using the different bosonized actions obtained in the preceding one.

Since the coupling constant in eq. (2.1) is dimensionless the theory is renormalizable and infinities occurring when computing quantum corrections can be elimi-

nated within a given renormalization scheme. Different prescriptions can be used, following for example refs. [13, 27–30]. We shall follow the approach described in ref. [30] in which the background-field method is employed to study coupling-constant renormalization (see appendix C for details).

We write the fields h and l introduced in sect. 2 in terms of backgrounds $\bar{h}$ and $\bar{l}$ and fluctuations π , φ

$$h(x) = \bar{h}(x) e^{i\pi(x)}, \quad l(x) = \bar{l}(x) e^{i\varphi(x)}, \quad (3.1)$$

where $\pi = \pi^a t^a$, $\varphi = \varphi^a t^a$ and t^a are the $SU(N)$ generators normalized according to eq. (2.4)

$$\begin{aligned} \text{tr } t^a t^b &= \tfrac{1}{2} \delta^{ab}, & f^{abc} f^{a'bc} &= \mathbb{C}(G) \delta^{aa'}, \\ [t^a, t^b] &= i f^{abc} t^c, & t^a t^a &= \mathbb{C}_d I. \end{aligned} \quad (3.2)$$

Divergencies of the theory are contained in the generating functional

$$Z[\varphi, \pi] = \int \mathcal{D}\varphi \mathcal{D}\pi \exp[iS(\bar{h} e^{i\pi}, \bar{l} e^{i\varphi}) - iS(\bar{h}, \bar{l}) - (\text{terms linear in } \varphi \text{ and } \pi)], \quad (3.3)$$

with

$$S(h, l) = (1 + 2N)W(h) + S_{\text{int}}(h, l) + S_{\text{impurity}}(l). \quad (3.4)$$

The Wess–Zumino–Witten action in eq. (3.4) is adjusted so that conformal invariance is respected [14–31] and hence the contribution of W to the β -function will vanish (see appendix C). Concerning the impurity action term, in both the standard non-abelian bosonization and the Wilson loop result, it gives no contribution to the β -function. Indeed, in the former case the β -function is zero for the same reasons we pointed out that concern the Wess–Zumino–Witten action. In the Wilson loop case, the impurity action term being $\det(1 + l(T)l^{-1}(0))$, a local operator average, it cannot contribute to the β -function. Hence, only the interaction term is responsible of coupling-constant renormalizations. Now, in both approaches, the action term describing the interaction between electrons and the impurity is given by

$$S_{\text{int}}(h, l) = \frac{1}{g} \text{tr} \int d^2x (ih^{-1} \partial_+ h + F)(il^{-1} \partial_l l + H). \quad (3.5)$$

Within the background-field method, one starts by dressing the fields $\bar{h}$ and $\bar{l}$ with quantum fluctuations $e^{i\varphi}$ and $e^{i\pi}$ according to eq. (3.1). Counterterms can then be

calculated using the relation

$$S_{\text{int}}(\bar{h} e^{i\varphi}, \bar{l} e^{i\pi}) - S_{\text{int}}(\bar{h}, \bar{l}) = \int_0^1 dt \frac{d}{dt} [S_{\text{int}}(\bar{h} e^{it\pi}, \bar{l} e^{it\pi})]. \quad (3.6)$$

One then has

$$\frac{1}{g^2} \text{tr} \int d^d x d^d y \varphi(x) (\bar{l}^{-1} \partial_t \bar{l} - 1) \partial_x \pi(x) \varphi(y) (\bar{l}^{-1} \partial_t \bar{l} + \bar{h}^{-1} \partial_+ \bar{h} + 1) \partial_t \pi(y), \quad (3.7)$$

or, Fourier transforming,

$$\begin{aligned} \delta S_{\text{int}} &= \frac{1}{g^2} \text{tr} \int \frac{d^2 k d^2 k'}{(2\pi)^4} [(\bar{l}^{-1} \partial_t \bar{l})(\bar{h}^{-1} \partial_+ \bar{h})]_{(-k-k', k+k')}^{ijpm} \\ &\quad \times k'_1 (-k'_0) \tilde{\Delta}_{ip}^{\varphi}(k) \tilde{\Delta}_{jm}^{\pi}(k'), \end{aligned} \quad (3.8)$$

where

$$\begin{aligned} \langle \tilde{\varphi}_i(k) \tilde{\varphi}_p(q) \rangle_0 &= i \tilde{\Delta}_{ip}^{\varphi}(k, q) = -(2\pi)^2 \delta^{(2)}(k+q) \tilde{\Delta}_{ip}^{\varphi}(k), \\ \langle \tilde{\pi}_j(k') \tilde{\pi}_m(q) \rangle_0 &= i \tilde{\Delta}_{jm}^{\pi}(k', q) = -(2\pi)^2 \delta^{(2)}(k'+q) \tilde{\Delta}_{jm}^{\pi}(k'), \end{aligned} \quad (3.9)$$

$$\begin{aligned} \tilde{\Delta}_{ip}^{\varphi}(k) &= -\frac{4\pi}{1+2N} \frac{\delta_{ip}}{k_0 k_1}, \\ \tilde{\Delta}_{jm}^{\pi}(k') &= -\frac{4\pi}{1+2N} \frac{\delta_{jm}}{k'^2 + m^2}. \end{aligned} \quad (3.10)$$

After some calculations one obtains

$$\delta S_{\text{int}} = \frac{1}{g^2 \pi^2} \frac{\text{tr}}{(1+2N)^2} (2\pi)^2 \int d^2 x (\bar{l}^{-1} \partial_t \bar{l})(x) (\bar{h}^{-1} \partial_+ \bar{h})(x) t^i t^j t^i t^j \int \frac{d^2 k}{k^2 + m^2}. \quad (3.11)$$

We employ dimensional regularization to tame ultraviolet divergencies. The divergent counterterm then results ($\varepsilon = 2 - d$)

$$\delta S_{\text{int}} = -\frac{2\pi}{g^2 (1+2N)^2} \frac{N^2 - 1}{N\varepsilon} \int d^d x (\bar{l}^{-1} \partial_t \bar{l})_{(x)} (\bar{h}^{-1} \partial_+ \bar{h})_{(x)}. \quad (3.12)$$

One then needs a coupling-constant renormalization. Calling μ the mass scale

introduced in the renormalization procedure and g_0 and g the bare and renormalized coupling constants

$$\frac{1}{g_0} = Z\mu^\epsilon \frac{1}{g}, \quad (3.13)$$

one then has

$$g\beta\left(\frac{1}{g}\right) + \beta\left(\frac{1}{g}\right) \frac{\partial \ln z}{\partial (1/g)} - \epsilon = 0, \quad (3.14)$$

with

$$\beta\left(\frac{1}{g}\right) = \mu \frac{\partial (1/g)}{\partial \mu} \Big|_{g_0, \Lambda}. \quad (3.15)$$

We then find that

$$\beta\left(\frac{1}{g}\right) = -\frac{\epsilon}{g + \partial \ln z / \partial (1/g)}. \quad (3.16)$$

Now, from eq. (3.12), we have

$$z = 1 - \frac{A}{g} \frac{1}{\epsilon}, \quad (3.17)$$

with

$$A = \frac{8\pi}{(1+2N)^2} \left[-\frac{N^2-1}{4N} \right], \quad (3.18)$$

so that

$$\beta(1/g) = -\frac{\epsilon}{g - A/\epsilon}. \quad (3.19)$$

If we consider the strong-coupling limit ($g \rightarrow \infty$) and let $\epsilon \rightarrow 0$, we obtain

$$\beta(g) = -\frac{2\pi}{(1+2N)^2} \frac{N^2-1}{N} + O\left(\frac{1}{g}\right). \quad (3.20)$$

One then discovers the existence of an infrared-stable fixed point (IRSFP) at $g \rightarrow \infty$. The sign of the β -function for large g is consistent with the standard perturbation calculation for the β -function for small g [27]. The IRSFP at infinity represents the ground state with a screened impurity. The β -function calculated above is valid for both the standard non-abelian bosonization and the Wilson-loop approaches.

4. Magnetic susceptibility and low-temperature specific heat

In this section we compute the magnetic susceptibility and the (low-temperature) specific heat of the Kondo model using the functional methods developed in sects. 1, 2 and 3. In particular, we study these physical quantities at the infrared stable fixed-point, i.e. the strong-coupling limit. As a bonus, we find the correlation functions at strong coupling. We also show that at any non-zero temperature there are no phase transitions thus overcoming one of the main limitations of the large degeneracy approach (see ref. [6]).

4.1. MAGNETIC SUSCEPTIBILITY

The Zeeman coupling of the system to an external magnetic field has the form

$$H^{\text{ext}} = \int d^3x c_\alpha^\dagger(x) \sigma_{\alpha\beta} c_\beta(x) \cdot \mathbf{H} + \mathbf{H} \cdot \mathbf{S} \quad (4.1)$$

where the first term is the Zeeman coupling of the band electrons and the second is the coupling of the impurity to the external field. In the notation of sect. 2 these terms play the same role as the coupling to external sources with $F = H = \mathbf{H} \cdot \boldsymbol{\sigma}$, with F and H in the algebra of $\text{SU}(N)$. It is shown in sect. 2 that the generating functional can be written in the form

$$Z[H] = \int \mathcal{D}A_+ \mathcal{D}A_- \hat{P} \exp \left\{ \frac{i}{g} \int d^2x (A_+ - H)(A_- - H) \right\} \\ \times \det(i\partial_+ + A_+) \det(i\partial_- + A_-), \quad (4.2)$$

with $\hat{P}$ a projector onto singly occupied impurity states (as in sect. 2)

The local magnetization $M(x)$ is

$$M^a(x) = \frac{1}{iZ} \frac{\delta Z}{\delta H^a(x)}. \quad (4.3)$$

Thus, we have

$$M^a(x) = \frac{1}{g} H^a(x) - \frac{1}{g} \langle A_+^a(x) + A_-^a(x) \rangle. \quad (4.4)$$

In terms of the Wess–Zumino–Witten fields we get

$$M^a(x) = \frac{1}{g} H^a(x) + \frac{i}{g} \text{tr} \left[\langle h^{-1}(x) \partial_+ h(x) + l^{-1}(x) \partial_- l(x) \rangle t^a \right]. \quad (4.5)$$

Analogously, the correlation function $\chi^{ab}(x, x')$, defined by

$$\chi^{ab}(x, x') = \frac{1}{iZ} \frac{\delta^2 Z}{\delta H^a(x) \delta H^b(x')} \bigg|_{H=0}, \quad (4.6)$$

is equal to

$$\begin{aligned} \chi^{ab}(x, x') &= \frac{1}{g} \delta(x - x') \delta^{ab} \\ &- \frac{i}{g} \left\langle \text{tr} \left[(h^{-1}(x) \partial_+ h(x) + l^{-1}(x) \partial_t l(x)) t^a \right] \right. \\ &\quad \left. \times \text{tr} \left[(h^{-1}(x') \partial_+ h(x') + l^{-1}(x') \partial_t l(x')) t^b \right] \right\rangle. \end{aligned} \quad (4.7)$$

The zero-field magnetic susceptibility is then given by

$$\chi^{ab} = \int d^2x \frac{1}{iZ} \frac{\delta^2 Z}{\delta H^a(x) \delta H^b(0)}. \quad (4.8)$$

This is the *total susceptibility* of the system. The Kondo susceptibility χ_{Kondo} is found by subtracting-off the free-fermion susceptibility

$$\chi_{\text{Kondo}} = \chi - \chi(g=0). \quad (4.9)$$

We will consider only the strong-coupling limit. In this limit, the system is scale invariant (since $\beta(1/g) \xrightarrow{g \rightarrow \infty} 0$) and the resulting conformal invariance allows for a detailed description of this limit. A systematic perturbation expansion in powers of $1/g$ is possible. However, we will not perform such an expansion here.

In the strong-coupling limit the h and l fields decouple (cf. eq. (2.25)). Thus

$$\langle h^{-1} \partial_+ h t^a \rangle_{g=\infty} = \langle l^{-1} \partial_t l t^a \rangle_{g=\infty} = 0. \quad (4.10)$$

The correlation functions will be calculated in the limit $g \rightarrow \infty$ using the methods of sect. 2 (A) (truly correct if k is large or, equivalently, if N is large). In this limit we can use the approximations

$$\begin{aligned} h &\approx 1 + i\Pi^a t^a, \\ l &\approx 1 + i\varphi^a t^a. \end{aligned} \quad (4.11)$$

Thus

$$\begin{aligned} \text{tr}(h^{-1} \partial_+ h t^a) &\approx \frac{1}{2} i \partial_+ \Pi^a, \\ \text{tr}(l^{-1} \partial_t l t^a) &\approx \frac{1}{2} i \partial_t \varphi^a. \end{aligned} \quad (4.12)$$

To leading order, the real-time correlation function is given by

$$\chi^{ab}(x, x') = \frac{1}{g} \delta^{ab} \delta(x - x') + \frac{i}{4g^2} \left[\langle \partial_+ \Pi^a(x) \partial_+ \Pi^b(x') \rangle + \langle \partial_t \varphi^a(x) \partial_t \varphi^b(x') \rangle \right. \\ \left. + \langle \partial_+ \Pi^a(x) \partial_t \varphi^b(x') \rangle + \langle \partial_t \varphi^a(x) \partial_+ \Pi^b(x') \rangle \right]. \quad (4.13)$$

The correlation functions inside the square brackets are

$$\langle \Pi^a(x) \Pi^b(x') \rangle = -i\delta^{ab} \int \frac{d^2k}{(2\pi)^2} \left(\frac{2N+1}{\pi} \right) \frac{k_0 k_1}{\mathcal{D}} e^{ik \cdot (x-x')}, \quad (4.14)$$

$$\langle \varphi^a(x) \varphi^b(x') \rangle = -i\delta^{ab} \int \frac{d^2k}{(2\pi)^2} \left(\frac{2N+1}{8\pi} \right) \frac{k^2 k_0^2}{\mathcal{D}} e^{ik \cdot (x-x')}, \quad (4.15)$$

$$\langle \Pi^a(x) \varphi^b(x') \rangle = \frac{i\delta^{ab}}{2g} \int \frac{d^2k}{(2\pi)^2} \frac{k_+ k_0}{\mathcal{D}} e^{ik \cdot (x-x')}, \quad (4.16)$$

where

$$\mathcal{D} = \left(\frac{2N+1}{8\pi} \right)^2 k^2 k_0 k_1 - \frac{k_1^2 k_0^2}{4g^2}, \\ k^2 = k_0^2 - k_1^2, \quad k_{\pm} = k_0 \pm k_1, \\ k \cdot x = k_0 x_0 - k_1 x_1 = k_+ x_- + k_- x_+. \quad (4.17)$$

By use of these results we find for the Fourier transform of $\chi^{ab}(x, x')$,

$$\chi^{ab}(k) = \delta^{ab} \chi(k), \quad (4.18)$$

where

$$\chi(k) = \frac{1}{g} + \frac{1}{g^2 \mathcal{D}} \left(\frac{2N+1}{32\pi} \right) (k_0 k_1 k_-^2 + k^2 k_0^2) - \frac{1}{8g} k_+ k_- k_0^2. \quad (4.19)$$

Here we are interested in the *static susceptibility* defined by the limit

$$\chi_{dc} = \lim_{k_1 \rightarrow 0} \lim_{k_0 \rightarrow 0} \chi(k). \quad (4.20)$$

We find

$$\chi_{dc} = \frac{1}{g} - \frac{1}{g^2} \frac{2\pi}{2N+1} + O\left(\frac{1}{g^3}\right). \quad (4.21)$$

Conversely, the zero-momentum limit of the dynamic correlation function $\chi(k_0, 0)$ is given by

$$\chi(k_0, 0) \simeq \frac{3}{2g} - \frac{2N+1}{8\pi}, \quad (4.22)$$

which does not appear to have a smooth $1/g$ expansion.

4.2. SPECIFIC HEAT

The specific heat can be calculated by writing the path-integral representation of the partition function Z ,

$$Z = \text{tr} e^{-\beta H}. \quad (4.23)$$

This is exactly the path-integral discussed in sects. 2 and 3, in imaginary time. In terms of the Wess–Zumino–Witten variables it takes the form

$$Z = \sqrt{\det(i\hat{\phi})} \det(i\partial_t) \int \mathcal{D}h \mathcal{D}l \hat{P} \exp \left[-kS_{\text{WZ}}(h) - kS_{\text{WZ}}(l) - \frac{1}{g} \int A_+ A_- \right], \quad (4.24)$$

where, once again, $\hat{P}$ is a projector.

A one-dimensional quantum system at non-zero temperature T ($T = 1/\beta$) can be treated performing the path-integral on a strip of length L (the size of the system) and width β , using periodic boundary conditions.

At $g = \infty$ we can write

$$Z = Z_{\text{free}}^0 Z_{\text{spin}}^0 Z_{\text{WZ}} Z_{\text{WZ}}, \quad (4.25)$$

where Z_{free}^0 is the partition function for free right-moving fermions (the band electrons), Z_{spin}^0 is the partition function of a free spin and Z_{WZ} and Z_{WZ} represent the partition functions of the fields h and l , which decouple as $g \rightarrow \infty$.

Now, Z_{WZ} is independent of the temperature. This is so (in all the three representations discussed in sect. 2) because the action S_{WZ} is independent of the metric. Since a change in temperature $T \rightarrow T'(\beta \rightarrow \beta')$ is equivalent to a dilatation of the imaginary time scale (and hence a change in metric), independence of the metric implies independence of the temperature. Thus, the only temperature dependence in Z is present through either the free-fermion or the Wess–Zumino part.

Thus, the specific heat of the impurity is entirely determined by Z_{WZ} alone. At $g = \infty$ the Kondo problem is scale invariant ($\beta(1/g) = 0$). Since any one-dimensional scale-invariant system is also conformally invariant (see refs. [32, 33]) we can use the results of Affleck [34] and Cardy et al. [35] to write down the low-temperature specific heat for the impurity in terms of the conformal anomaly c of the level

k Wess–Zumino–Witten model

$$c_{\text{imp}} = \frac{1}{6} \pi c T, \quad (4.26)$$

with

$$c = \frac{k \dim G}{k + \mathfrak{C}(G)}, \quad (4.27)$$

for a Lie group G . For $SU(N)$ we get

$$c = \frac{(2N+1)(N^2-1)}{2N+1+N} = \frac{(2N+1)(N^2-1)}{3N+1}. \quad (4.28)$$

5. Conclusions

We have presented a bosonized action for the $SU(N)$ Kondo problem using the method of path-integral non-abelian bosonization. In sect. 3 we derived the bosonized action and showed that it is equivalent to a level- $k = 2N + 1$ Wess–Zumino–Witten theory coupled to the impurities. We showed that the large- g limit is conformally invariant by explicit calculation of the β -function. We have calculated physical quantities, such as the static magnetic susceptibility and the low-temperature specific heat, in the large- g limit. We have not attempted to study the Kondo crossover within this approach. It is worth noting that, contrary to other path-integral approaches (such as the $1/N$ expansion), there are no phase transitions as functions of temperature in our approach.

Many issues remain unsolved. It would be nice to develop the geometric quantization approach (one of the three alternative bosonization schemes we presented) fully so as to make contact with the picture of Anderson, Yuval and Hamman (AYH) [1]. One virtue of our methods is that we never had to reduce the symmetry as in the AYH treatment.

Our approach puts the strong-coupling Kondo problem with other one-dimensional scale-invariant theories. It appears natural to use the methods of conformal field theory to the study of the strong-coupling limit. Finally, our approach appears to be naturally suited for a semiclassical study of the Kondo problem. In particular, it would be interesting to identify the fractionally charged solutions present in the Bethe ansatz with solutions in our bosonized action.

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Appendix A

We first show in this appendix the relation between the fermion determinant in the fundamental representation and the jacobian relating $\mathcal{D}A_+$ with $\mathcal{D}h$ [14, 19–21]

$$\mathcal{D}A_+ = J(h) \mathcal{D}h. \quad (\text{A.1})$$

From the definition

$$A_+ = -\frac{i}{g} h^{-1} \partial_+ h, \quad (\text{A.2})$$

it is easy to see that

$$J(h) = \det(i\partial_+ - g[A_+, \cdot]) = \det D_+^{\text{Adj}}[A], \quad (\text{A.3})$$

$D_+^{\text{Adj}}[A]$ being an unbounded operator, and $J(h)$ an ill-defined quantity which requires some regularization (formal manipulations, such as those presented in ref. [36] leading to a trivial value for $J(h)$, pass over this problem). Following refs. [14, 19–21] we adopt the same regularization for eq. (A.3) as that used for the fermion determinant in the fundamental representation (i.e. heat-kernel or zeta-function methods). Now, it is easy to show that

$$\det \frac{D_+^{\text{Adj}}}{i\partial_+} = \left[\det \frac{D_+^{\text{Fund}}}{i\partial_+} \right]^{2\mathbb{C}(G)} \quad (\text{A.4})$$

with $\mathbb{C}(G)$ the quadratic Casimir in the adjoint representation. This relation is due to the fact that the anomaly $\mathcal{A}$, in any representation, is proportional to $\text{tr } T^a T^b$ with T^a the generators in the corresponding representation. Now, the logarithm of fermion determinants like those given in eq. (A.4) can be obtained from the anomaly [15–17] by integrating over a parameter t which interpolates between $h(x, t=1) = h(x)$ and $h(x, t=0) = I$. Then, eq. (A.4) follows after use of the relation

$$\text{tr } T_{\text{Adj}}^a T_{\text{Adj}}^b = 2\mathbb{C}(G) \text{tr } T_{\text{Fund}}^a T_{\text{Fund}}^b. \quad (\text{A.5})$$

Concerning the fermion determinant in the fundamental representation, fermions being left-handed, the corresponding Dirac operator has to be modified by addition of free right-handed fermions [17]. One can then integrate the anomaly [16] to obtain

$$\det D_+ \equiv \det \hat{D}|_{\text{Reg}} = \exp \left[iW[h] + i\alpha \text{tr} \int h^{-1} \partial_+ h \cdot h^{-1} \partial_- h \right]. \quad (\text{A.6})$$

Here α is an arbitrary parameter arising from regularization ambiguities. Conformal invariance dictates $\alpha = 0$ [31].

Appendix B

In this appendix we compute the determinant

$$J_F = \frac{\det(i\partial_t + H(t))}{\det(i\partial_t)}, \quad (\text{B.1})$$

following the technique developed in ref. [37].

Consider the generating functional

$$Z[H] = \lim_{T \rightarrow \infty} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[i \int_0^T dt \bar{\psi}(t) (i\partial_t + H(t)) \psi(t) \right], \quad (\text{B.2})$$

where $\bar{\psi}$, ψ are Grassmann fields and $H(t)$ is in the algebra of $U(N)$.

Then, we find

$$J_F = \det(i\partial_t) Z[H]. \quad (\text{B.3})$$

We can compute Z quite explicitly. We use a gauge-invariant regulator (Pauli-Villars) and pick the gauge $H(t) = \bar{H}$, with

$$T\bar{H} = \int_0^T dt H(t). \quad (\text{B.4})$$

Under gauge transformations $U(t) \in U(N)$, $\psi(t)$ and $H(t)$ transform like

$$\begin{aligned} \psi(t) &\rightarrow U(t)\psi(t) = \psi'(t), \\ H(t) &\rightarrow U^{-1}(t)H(t)U(t) + iU^{-2}\partial_t U = H'(t). \end{aligned} \quad (\text{B.5})$$

The object

$$W = \hat{T} e^{i \int_0^T dt H(t)}, \quad (\text{B.6})$$

(where $\hat{T}$ stands for a time-ordered product) transforms like a group element

$$W \rightarrow U^{-1}(0) W U(T) = W'. \quad (\text{B.7})$$

If we pick periodic boundary conditions for the gauge fields $H(t)$, we have $U(0) = U(T)$. Thus, we can write

$$Z = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[i \int_0^T dt \bar{\psi}(t) (i\partial_t + \bar{H}) \psi(t) \right]. \quad (\text{B.8})$$

In euclidean space (imaginary time) we have

$$t = i\tau, \quad T = -i\beta \quad \text{and} \quad \bar{H} \equiv i\bar{H}.$$

Hence

$$Z = \int \mathcal{D}\bar{\psi}(\tau) \mathcal{D}\psi(\tau) \exp \left[- \int_0^T d\tau \bar{\psi}(\tau) (\partial_\tau + i\bar{H}) \psi(\tau) \right], \quad (\text{B.9})$$

with $\psi(\tau + \beta) = -\psi(\tau)$. By expanding in normal modes we find

$$\psi(\tau) = \sum_{n=-N_1}^{N_2} \psi_n e^{i(n+1/2)2\pi\tau/\beta}, \quad (\text{B.10})$$

where N_1 and N_2 are cutoffs. The action is

$$S = i \sum_{n=-N_1}^{N_2} \bar{\psi}_n \left[2\pi \left(n + \frac{1}{2} \right) + \beta \bar{H} \right] \psi_n. \quad (\text{B.11})$$

Elitzur et al. [37] use gauge invariant Pauli–Villars regulators and find

$$Z = \prod_{j=1}^k (1 + e^{i2\pi\bar{h}_j}), \quad (\text{B.12})$$

where $\{\bar{h}_j\}$ is the set of eigenvalues of the rank- k matrix $\beta\bar{H}/2\pi$. Clearly

$$Z = \det(1 + e^{i2\pi h}) \equiv \det(1 + e^{i\beta\bar{H}}). \quad (\text{B.13})$$

A rotation back to Minkowski space yields

$$Z = \det(1 + \hat{T} e^{i\int_0^T dt H(t)}) \equiv \det(1 + W) = \Delta, \quad (\text{B.14})$$

which is *independent of the representation*. An alternative derivation given by Blackenbecker et al. [22] leads to the same result.

Next, we consider the relation between fundamental and adjoint representations of $\text{SU}(N)$. Let us start by the finite determinant in the fundamental representation of $\text{SU}(N)$

$$\Delta_F = \det(1 + e^{iW \cdot t_F}), \quad (\text{B.15})$$

where t_F^a are the generators of $\text{SU}(N)$ in the fundamental representation (see eq. (2.4)). Using the invariance properties of the determinant, as well as the fact that it is a class function, we can write (for an arbitrary $U \in \text{SU}(N)$)

$$\Delta_F = \det[U^{-1}(1 + e^{iW \cdot t_F})U] = \det[1 + U^{-1}e^{iW \cdot t_F}U]. \quad (\text{B.16})$$

It is always possible to choose U such that $U^{-1}e^{iW \cdot t_F}U$ is a diagonal matrix, i.e.

$U^{-1} e^{i\tilde{W} \cdot t_F} U = e^{i\tilde{W} \cdot t_F}$, where $\tilde{W}$ has components only in the directions of the Cartan subalgebra $\{H_i^F\}$ with $i = 1, 2, \dots, N-1$. The diagonal generators $\{H_i^F\}$ have N eigenvectors $|\mu_F\rangle$ with eigenvalues $\mu_{i,F}^a$. The vectors μ_F^a are the weights of the fundamental representation of $SU(N)$ and satisfy

$$\sum_{a=1}^N \mu_F^a = 0. \quad (\text{B.17})$$

Thus

$$\Delta_F = \prod_{a=1}^N \left(1 + \exp \left(i \sum_{j=1}^r \tilde{W}_j \mu_{i_F}^a \right) \right). \quad (\text{B.18})$$

In the case of the adjoint representation we have to compute

$$\Delta_A = \det(1 + e^{i\tilde{W} \cdot t_A}), \quad (\text{B.19})$$

where t_A are the $SU(N)$ generators in the adjoint representation. Note that the parameters $\tilde{W}$ are the same as before. The same arguments used for the fundamental representation apply here and imply that Δ_A can be written in terms of the weights of the adjoint representation, i.e. the roots of the algebra. Now, the roots are either zero ($N-1$ of them) or differences of weights of the fundamental representation. Thus, we can write

$$\begin{aligned} \Delta_A &= \prod_{a=1}^{N^2-1} (1 + e^{i\tilde{W}_i H_i^a}) \\ &= 2^{N-1} \prod_{a \neq b=1}^N [1 + \exp(i\tilde{W}_i (\mu_{i_F}^a - \mu_{i_F}^b))]. \end{aligned} \quad (\text{B.20})$$

Using eq. (B.3) we get

$$\Delta_A = 2^{N-1} \prod_{a \neq b=1}^N (e^{i\tilde{W}_i \mu_{i_F}^a} + e^{i\tilde{W}_i \mu_{i_F}^b}). \quad (\text{B.21})$$

In the case of $SU(2)$ these expressions yield

$$\Delta_F = (1 + e^{i\tilde{W}/2})(1 + e^{-i\tilde{W}/2}) = 2 + 2 \cos \frac{1}{2} \tilde{W}, \quad (\text{B.22})$$

for the fundamental and

$$\begin{aligned} \Delta_A &= 2(e^{i\tilde{W}/2} + e^{-i\tilde{W}/2})(e^{-i\tilde{W}/2} + e^{i\tilde{W}/2}) \\ &= 8 \cos^2 \frac{1}{2} \tilde{W}, \end{aligned} \quad (\text{B.23})$$

for the adjoint representations. Thus, we find that

$$\Delta_A = 2(\Delta_F - 2)^2. \quad (\text{B.24})$$

Notice that this relation differs from the analogous result for the fermion determinants of Appendix A.

Appendix C

In this appendix we sketch the renormalization of the the non-linear sigma model with Wess–Zumino term using dimensional regularization. Following ref. [30], β -functions discussed in this paper can then be very easily computed.

Consider a scalar field $U(x)$, taking values in a simple Lie group, with dynamics defined by the two-dimensional action

$$W(U) = \frac{1}{4\lambda^2} \int d^2x \operatorname{tr} \partial_\mu U \partial^\mu U^{-1} + k\Gamma(U), \quad (\text{C.1})$$

with

$$\Gamma(U) = \epsilon^{ijk} \int d^3x (U^{-1} \partial_i U) (U^{-1} \partial_j U) (U^{-1} \partial_k U). \quad (\text{C.2})$$

As is well-known [13], conformal invariance implies

$$\lambda^2 = 4\pi/k. \quad (\text{C.3})$$

The renormalization of the non-linear sigma model defined in eq. (C.1) will be carried out within the background-field method. We write the field $U(x)$ as a background dressed with quantum fluctuations

$$U(x) = \bar{U}(x) e^{i\Pi(x)}, \quad \Pi(x) = \Pi^i(x) T^i, \quad (\text{C.4})$$

where T^i are generators of $\text{SU}(N)$. The divergences of the theory are contained in the functional

$$Z[\bar{U}] = \int \mathcal{D}\Pi \exp i [W(\bar{U} e^{i\Pi}) - W(\bar{U}) - (\text{terms linear in } \Pi)]. \quad (\text{C.5})$$

Using well-known properties of the Wess–Zumino functional Γ , we may write

$$\begin{aligned} W(\bar{U} e^{i\Pi}) - W(\bar{U}) &= \int_0^1 dt \frac{d}{dt} W(\bar{U} e^{it\Pi}) \\ &= \int d^2x \int_0^1 dt \frac{(-i)}{2\lambda^2} P^{\mu\nu} \operatorname{tr} \partial_\mu \Pi \{ e^{-it\Pi} \partial_\nu e^{it\Pi} + e^{-it\Pi} \bar{U}^{-1} \partial_\nu \bar{U} e^{it\Pi} \}. \end{aligned} \quad (\text{C.6})$$

where $P^{\mu\nu}$ is the tensorial expression $\eta^{\mu\nu} - k\lambda^2\epsilon^{\mu\nu}/4\pi$. To tame infrared divergencies we add a mass term to the action

$$W_m(U) = \frac{m^2}{4\lambda^2} \int d^2x \operatorname{tr}(U + U^\dagger). \quad (\text{C.7})$$

Ultraviolet regularization is carried out in a dimensional scheme. The regularized exponent in eq. (C.6) is

$$\begin{aligned} \int d^d x \left[\int_0^1 dt \frac{(-i)}{2\lambda^2} P^{\mu\nu} \operatorname{tr} \partial_\mu \Pi \left[e^{-it\Pi} \partial_\nu e^{it\Pi} + e^{-it\Pi} \bar{U}^{-1} \partial_\nu \bar{U} e^{it\Pi} \right] \right. \\ \left. + \frac{m^2}{4\lambda^2} \operatorname{tr} \left[\bar{U} e^{i\Pi} - \bar{U} + e^{-i\Pi} \bar{U}^\dagger - \bar{U}^\dagger - (\text{terms linear in } \Pi) \right] \right]. \quad (\text{C.8}) \end{aligned}$$

For the one-loop renormalization we need the part of eq. (C.8) quadratic in Π (we call this part $W_{\text{eff}}(\Pi)$)

$$\begin{aligned} W_{\text{eff}}(\Pi) = \frac{1}{2\lambda^2} \int d^d x \left(\partial_\mu \Pi^i \partial^\mu \Pi^i - m^2 \Pi^i \Pi^i \right) - \frac{i}{\lambda^2} \int d^d x f_{ijk} \partial_\mu \Pi^i \Pi^j P^{\mu\nu} (\bar{U}^{-1} \partial_\nu \bar{U})^k \\ - \frac{m^2}{8\lambda^2} \int d^d x \operatorname{tr} (\bar{U} + \bar{U}^\dagger) T^i T^j \Pi^i \Pi^j. \quad (\text{C.9}) \end{aligned}$$

In order to calculate counterterms we consider the second and third terms in W_{eff} as interaction terms, $\int d^d x \mathcal{L}_{\text{int}}(x)$, modifying the free (first) lagrangian terms. Then, expanding the interaction exponential in a power series, we can evaluate the integrals in increasing order of loop number.

There are two divergent one-loop diagrams. The renormalization of the couplings is determined by one-loop diagrams with two insertions of the background field $P^{\mu\nu} \bar{U}^{-1} \partial_\nu \bar{U}$. Its divergent part is

$$- \frac{iC_a}{8\pi\epsilon} \eta_{\mu\rho} P^{\mu\nu} P^{\rho\sigma} \int d^d x \operatorname{tr} \partial_\nu \bar{U} \partial_\sigma \bar{U}^{-1}, \quad (\text{C.10})$$

where $\epsilon = 2 - d$ and $\eta_{\mu\rho}$ is a d -dimensional metric tensor. Introducing a two-dimensional metric tensor $\bar{\eta}^{\mu\nu}$ defined through

$$\epsilon^{\mu\nu} \epsilon^{\rho\sigma} = \bar{\eta}^{\mu\sigma} \bar{\eta}^{\nu\rho} - \bar{\eta}^{\mu\rho} \bar{\eta}^{\nu\sigma},$$

we can define $\hat{\eta}^{\mu\nu} = \eta^{\mu\nu} - \bar{\eta}^{\mu\nu}$, which has the property $\hat{\eta}^\mu_\mu = d - 2$.

Minimal subtraction leads to the following counterterms

$$\begin{aligned}
 (\Delta W)_{1a} = & \frac{\mathbf{C}_a}{8\pi\epsilon} \left[1 - \left(\frac{K\lambda^2}{4\pi} \right)^2 \right] \int d^d x \, \eta^{\mu\nu} \text{tr} \, \partial_\mu U \partial_\nu U^{-1} \\
 & + \frac{\mathbf{C}_a}{8\pi\epsilon} \left(\frac{k\lambda^2}{4\pi} \right)^2 \int d^d x \, \hat{\eta}^{\mu\nu} \text{tr} \, \partial_\mu U \partial_\nu U^{-1}.
 \end{aligned} \tag{C.11}$$

There are two relevant aspects to denote:

- (i) the corrections to λ vanishes if $\lambda^2 = 4\pi/k$;
- (ii) an unexpected new term has appeared, containing an operator that vanishes in the limit $d \rightarrow 2$ – consistency requires that it must be taken into account in higher loop calculations, where its effects will be non-vanishing.

Concerning mass renormalization, to one loop order it comes from a tadpole whose divergent part is cancelled by the counterterm

$$(\Delta W)_{1b} = \frac{m^2 \mathbf{C}_d}{16\pi\epsilon} \int d^d x \, \text{tr}(U + U^\dagger). \tag{C.12}$$

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